

$$\begin{aligned}
C_{\text{dw}}^{11,12} \rightarrow \frac{1}{16\pi^2} & \left(-\frac{1}{48} g_2 c_\gamma \frac{y_d^{\text{pi}2}}{m_\phi^2} (s_\gamma^2 \overline{y_d^{\text{pr}}} (y_d^{\text{pi}2} y_d^{\text{i}1r} + 18 y_d^{\text{pr}} y_d^{\text{i}12}) + y_d^{\text{pi}2} \overline{y_u^{\text{pr}}} y_u^{\text{i}1r} (c_\gamma^2 + 3 s_\gamma^2)) \right) + \\
& \frac{1}{8} g_2 c_\gamma s_\gamma^2 y_d^{\text{pi}2} \overline{y_u^{\text{pr}}} y_u^{\text{i}1r} \text{LF}_{1,2}[\mathbf{m}_\oplus] + \frac{1}{144} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}] - \\
& \frac{1}{72} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}] + \frac{1}{144} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}] - \\
& \frac{1}{16} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,0}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}] - \frac{1}{8} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,-1}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{3}{16} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{4,1,-2}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}] + \frac{1}{3} g_2 c_\gamma g_3^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,0}[\mathbf{m}_3, \mathbf{m}_q^{\text{i}1}] - \\
& \frac{2}{3} g_2 c_\gamma g_3^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,-1}[\mathbf{m}_3, \mathbf{m}_q^{\text{i}1}] + \frac{1}{3} g_2 c_\gamma g_3^2 y_d^{\text{i}1\text{i}2} \text{LF}_{4,1,-2}[\mathbf{m}_3, \mathbf{m}_q^{\text{i}1}] - \\
& \frac{1}{8} g_2 c_\gamma \overline{y_d^{\text{pr}}} y_d^{\text{pi}2} y_d^{\text{i}1r} \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_d^r] + \frac{1}{8} g_2 c_\gamma \overline{y_d^{\text{pr}}} y_d^{\text{pi}2} y_d^{\text{i}1r} \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_d^r] - \\
& \frac{1}{8} g_2 c_\gamma y_d^{\text{pi}2} \overline{y_u^{\text{pr}}} y_u^{\text{i}1r} \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_u^r] + \frac{1}{8} g_2 c_\gamma y_d^{\text{pi}2} \overline{y_u^{\text{pr}}} y_u^{\text{i}1r} \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_u^r] + \\
& \frac{1}{72} g_2 m_1 g_1^2 (c_\gamma a_d^{\text{i}1\text{i}2} - s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{144} g_2 m_1 g_1^2 (-c_\gamma a_d^{\text{i}1\text{i}2} + s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{72} g_2 m_1 g_1^2 (-c_\gamma a_d^{\text{i}1\text{i}2} + s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{144} g_2 m_1 g_1^2 (c_\gamma a_d^{\text{i}1\text{i}2} - s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \mathbf{m}_d^{\text{i}2}] + \\
& \frac{1}{24} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{1,1,1,0}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] - \frac{1}{12} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] - \\
& \frac{1}{24} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \frac{1}{24} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \\
& \frac{1}{24} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \frac{1}{24} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-2}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_d^{\text{i}2}] + \\
& \frac{1}{24} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_d^{\text{i}2}] + \frac{1}{48} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-2}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{48} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_q^{\text{i}1}] - \frac{1}{8} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{1,1,1,0}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] - \\
& \frac{1}{4} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,-1}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \frac{1}{8} m_2 s_\gamma \tilde{\mu} g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,0}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \\
& \frac{3}{8} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-2}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \frac{3}{8} m_2 s_\gamma \tilde{\mu} g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-1}[\mathbf{m}_2, \mathbf{m}_q^{\text{i}1}, \tilde{\mu}] + \\
& \frac{3}{16} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-2}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_q^{\text{i}1}] + \frac{3}{16} m_2 s_\gamma \tilde{\mu} g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,2,1,-1}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{3} g_2 m_3 g_3^2 (-c_\gamma a_d^{\text{i}1\text{i}2} + s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,1,1,0}[\mathbf{m}_3, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{6} g_2 m_3 g_3^2 (c_\gamma a_d^{\text{i}1\text{i}2} - s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,2,1,-1}[\mathbf{m}_3, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{3} g_2 m_3 g_3^2 (c_\gamma a_d^{\text{i}1\text{i}2} - s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{3,1,1,-1}[\mathbf{m}_3, \mathbf{m}_d^{\text{i}2}, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{6} g_2 m_3 g_3^2 (-c_\gamma a_d^{\text{i}1\text{i}2} + s_\gamma \tilde{\mu} y_d^{\text{i}1\text{i}2}) \text{LF}_{2,2,1,-1}[\mathbf{m}_3, \mathbf{m}_q^{\text{i}1}, \mathbf{m}_d^{\text{i}2}] + \\
& \frac{1}{16} g_2 \tilde{\mu} y_d^{\text{pi}2} y_u^{\text{i}1r} (-s_\gamma \overline{a_u^{\text{pr}}} + \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) \text{LF}_{2,2,1,-1}[\mathbf{m}_q^{\text{p}}, \tilde{\mu}, \mathbf{m}_u^r] + \\
& \frac{1}{16} g_2 \tilde{\mu} y_d^{\text{pi}2} y_u^{\text{i}1r} (s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) \text{LF}_{2,2,1,-1}[\mathbf{m}_u^r, \tilde{\mu}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{12} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,-1}[\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_d^{\text{i}2}] + \frac{1}{12} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-2}[\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_d^{\text{i}2}] + \\
& \frac{1}{12} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{3,1,1,-1}[\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_d^{\text{i}2}] - \\
& \frac{1}{24} g_2 c_\gamma g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,-1}[\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{\text{i}1}] - \frac{1}{24} g_2 m_1 s_\gamma \tilde{\mu} g_1^2 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,0}[\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{8} c_\gamma g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,-1}[\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{\text{i}1}] + \frac{1}{8} m_2 s_\gamma \tilde{\mu} g_2^3 y_d^{\text{i}1\text{i}2} \text{LF}_{2,1,1,0}[\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_q^{\text{i}1}] + \\
& \frac{1}{8} g_2 \tilde{\mu} y_d^{\text{pi}2} y_u^{\text{i}1r} (-s_\gamma \overline{a_u^{\text{pr}}} + \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) \text{LF}_{2,1,1,0}[\tilde{\mu}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_u^r] + \\
& \frac{1}{8} g_2 \tilde{\mu} y_d^{\text{pi}2} y_u^{\text{i}1r} (-s_\gamma \overline{a_u^{\text{pr}}} + \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}}) \text{LF}_{3,1,1,-1}[\tilde{\mu}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_u^r] \Big)
\end{aligned}$$